



Folded Twice, Told Anyway: A Quasi-Experimental Audit of Secrecy Leakage in Paper-Slip Versus App-Matched Office Gift Exchanges

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Abstract. Workplace gift exchanges increasingly delegate their anonymity to a matching app rather than a folded-slip hat draw, on the promise that an algorithm keeps a secret better than a hallway can. We report a quasi-experimental audit of twenty workplace teams (187 participants; eleven teams drew names on paper, nine through an app) running an annual gift exchange, tracking whether each participant's assigned recipient was known or strongly suspected before the reveal, through which channel any such leak reached them, and whether the team took corrective action once a leak was noticed. Overall leak incidence was modestly lower under the app (19.3% of participants) than on paper (24.0%), a gap not clearly distinguishable from chance. The leak channel, however, differed sharply: paper-slip leaks travelled mostly through direct exposure to the slip itself or a loose word from the organiser, while app-matched leaks travelled mostly through a lock-screen notification preview or a shared, still-logged-in device. Restricting the app sample to teams whose invite explicitly promised anonymity left the gap unchanged. App-matched teams were also markedly more likely to issue a corrective re-draw once a leak surfaced. We interpret this as channel substitution rather than secrecy improvement: formalising the draw relocates where a gift exchange leaks without obviously reducing how often it does.

Introduction

A hallway hat draw and a phone app promise the same thing — that nobody finds out who they're buying for until the day everyone opens their gift together — through very different mechanisms. The app's pitch is specifically that it does this better: assignments never touch a shared surface, nobody sees a slip, and the match lives inside an interface only its recipient is meant to open. Whether that promise changes the exchange's actual secrecy record, or simply moves the record's failure mode somewhere else, has not to our knowledge been audited directly.

We report a quasi-experimental comparison of twenty workplace teams running an annual Secret Santa exchange, split by the draw method each team's organiser had already chosen before we became involved. This design trades

randomisation for ecological validity: the teams in our sample used whichever method suited their office already, and we observed rather than assigned it.

Our contributions:

- We report a twenty-team, 187-participant field audit of leak incidence, leak channel, and post-leak containment behaviour under paper-slip and app-matched gift-exchange draws.
- We find the two methods' overall leak-incidence gap is modest and not clearly distinguishable from chance, though the channel through which a leak reaches its victim differs sharply between methods, and this pattern survives restricting the app sample to teams whose invite explicitly marketed the match as anonymous.
- We propose a channel-substitution account of formalised small-group secrecy tools, which

may generalise to other settings where a group replaces an informal information-withholding norm with a dedicated instrument.

Related work

Secrecy is effortful to hold rather than passively absent: unwanted mind-wandering back to a secret, not the act of concealing it in company, predicts the burden people report carrying [1], and thinking about a secret evokes a felt conflict between wanting to share it and needing not to [2] — a tension a gift-exchange participant carries for exactly as long as their assignment stays live. Gossip is not merely a leak vector but a bonding mechanism in its own right, and people are more motivated to pass on exactly the kind of norm-violating information a premature reveal represents [3]. Organisational rituals such as an annual gift exchange function to mark shared membership and manage low-grade anxiety around the calendar event they attach to [4], which is one reason a team keeps running the ritual even once its stated purpose (surprise) starts failing.

A second literature concerns what happens when a group's informal rule for managing a shared resource is replaced by a formal instrument. Formalising a previously informal arrangement does not always strengthen compliance with the arrangement's intent, and can crowd out the informal norm it was meant to support [5]; cooperation within a group is often sustained instead by evolved norms of reciprocity and reputation rather than by an externally imposed rule [6]. Elsewhere at this institution, formalising a shared household resource's ledger did not eliminate the informal workaround economy that had grown up around the resource it tracked [7], a pattern this study's channel data extends from a resource to a secret. Interface design itself is a lever on behaviour in exactly this way, steering choices through default and framing decisions a user rarely notices making [8]. Running the draw is also, structurally, a coordination task delegated to one member of the group, and effort and accountability for a shared outcome are well documented to diffuse across the rest of a group once responsibility for it is unclear [9]. We treat the paper-slip draw as the informal instrument and the matching app as its formal replacement, and ask which secrecy-relevant behaviour survives the substitution.

Method

We recruited twenty workplace teams running an internal, opt-in gift exchange during the same holiday period, through internal notice-boards and word of mouth across several organisations; team size ranged from six to fourteen ($M = 9.35$). Eleven teams' organiser ran a traditional paper-slip hat draw (names folded, drawn blind by each participant in turn); nine ran an anonymous app-matched draw, in which participants entered their name once and the app privately notified each of their assignment. Method was not randomly assigned — it was each organiser's standing practice — so we treat the comparison as quasi-experimental throughout.

Within one week of each team's reveal event, every participant ($n = 187$) completed a short anonymous post-event survey. Self-report of one's own information-handling is a well-studied assessment instrument in its own right, with well-documented reactivity effects in which the act of monitoring a behaviour changes the behaviour being monitored [10]; we return to this in Limitations. Because every participant also knew their team had opted into an audit, some part of any measured leak-avoidance behaviour may itself be a response to being watched rather than to the draw method, a confound with a long history in participant-observation research [11]. A participant was coded as having experienced a leak if their self-rated confidence c_i , on a 0–1 scale, that they could correctly name their assigned recipient **before** the reveal met a pre-registered threshold:

$$\text{leak}_i = \begin{cases} 1 & \text{if } c_i \geq 0.8 \\ 0 & \text{otherwise} \end{cases}$$

Participants who reported a leak were additionally asked to classify how the assignment reached them, in their own words; two coders independently sorted these open responses into three channels common across both methods — direct exposure (seeing the slip itself; a lock-screen notification preview), third-party disclosure (the organiser saying something; a shared, already-logged-in device), and contextual inference (deducing it from wrapping, timing, or a forwarded screenshot) — resolving disagreements by discussion (inter-coder agreement 91% before resolution). Organisers separately reported whether the team issued a corrective re-draw or reassignment once any leak became apparent to them. Group differences in propor-

tions were tested with χ^2 ; differences in the leak-to-reveal interval were tested with a Mann-Whitney U.

Results

Leak incidence was numerically lower under the app-matched draw (19.3%, 16/83 participants) than the paper-slip draw (24.0%, 25/104 participants), but the gap was not clearly distinguishable from chance ($\chi^2(1) = 0.62$, $p = .43$; Figure 1).

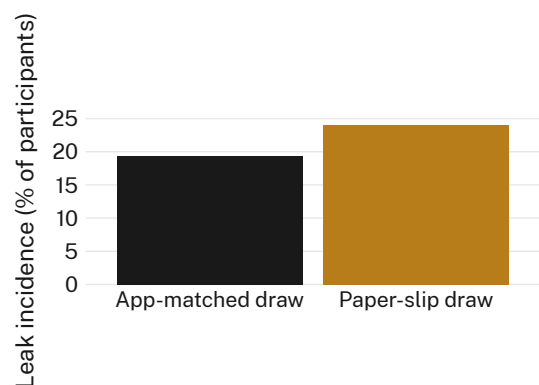


Figure 1: App-matched teams leaked to 19.3% of participants against paper-slip teams' 24.0% — a gap the sample cannot distinguish from chance.

Restricting the app-matched sample to the five teams whose invitation message explicitly promised “fully anonymous” matching left the rate essentially unmoved (18.8% marketed vs. 19.3% overall app-matched, $\Delta = 0.5$ percentage points, n.s.); we retain the full app-matched sample for completeness.

The channel a leak travelled through, in contrast, differed sharply by method (Figure 2). Among paper-slip leaks, 44% were direct exposure to the slip itself, 32% a loose word from the organiser, and 24% inferred from wrapping or timing. Among app-matched leaks, the direct-exposure share was, if anything, higher (50%), but its content changed entirely: every direct-exposure app leak traced to a lock-screen notification preview rather than a viewed slip, with the remainder split between a shared logged-in device (31%) and a forwarded screenshot (19%). A notification preview arriving unbidden is, functionally, a reminder in exactly the sense studied in the personal-savings literature, where a salient prompt raises the odds a piece of information reaches someone whether or not they went looking for it [12].

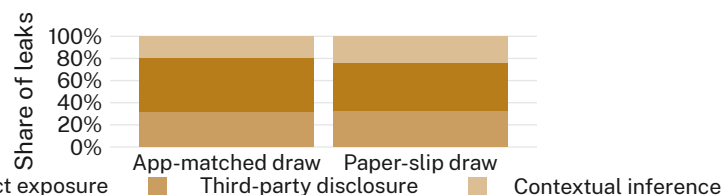


Figure 2: App-matched leaks reach their victim by roughly the same three-way split as paper-slip leaks, but every direct-exposure case is now a notification preview, not a viewed slip.

The leak-to-reveal interval showed a similar median under both methods (paper-slip 6 days, IQR 3–11; app-matched 5 days, IQR 2–9; Mann-Whitney U, $p = .61$; Figure 3), indicating that however a leak reaches its victim, it tends to surface on a comparable timescale before the reveal event under either method.

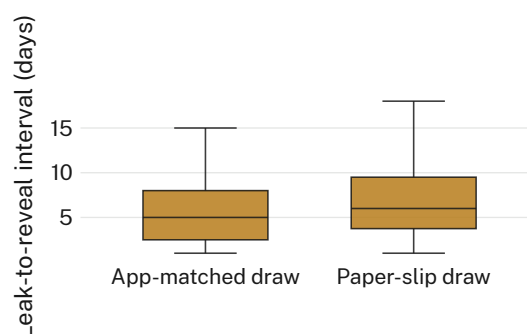


Figure 3: Days between a leak occurring and the scheduled reveal, by draw method — overlapping distributions under both methods.

Containment behaviour diverged more than incidence did. Of the 25 paper-slip leaks, two teams (8%) issued a corrective re-draw or re-assignment; of the 16 app-matched leaks, seven teams (44%) did. Organisers in app-matched teams described the re-draw itself as a low-friction action within the app; paper-slip organisers more often described simply hoping the recipient would “still act surprised” — a preference for repairing the group’s shared story over correcting the record that tracks a broader pattern in which visible cues about what a group is actually doing shape conduct more than a stated rule does [13], [14]. A visible, uncorrected leak may also invite a further one, echoing findings from physical public spaces that one piece of disorder left unaddressed measurably raises the rate of the next [15]. Notably, no organiser in either arm reported any sanction directed at whoever

caused a leak, despite evidence from public-goods settings that even a small, costly punishment option can materially raise a group's own cooperation rate [16] — the gift exchange's informal rules, whichever draw method carries them, apparently do not extend to policing violations of themselves. Prior work at this institution similarly found that making a metric newly visible did not change the underlying event it measured but shortened how quickly people responded to it once it happened [17], a response-side pattern this study's containment gap echoes on a different object entirely.

Discussion

Formalising a gift exchange's draw did not, in this sample, obviously improve on the paper slip it replaced at the outcome the app markets itself against: whether the secret holds. What the app changed was where a leak comes from and what a team does once it notices one. A lock-screen notification preview is, functionally, a new kind of loose word from the organiser — delivered by the device rather than the person, on the same rough timescale, at a similar overall rate. The sharper divergence was downstream of the leak: an app-matched team could re-draw with a few taps, and more of them did, while paper-slip teams more often let a spoiled surprise stand and improvised the reveal around it. A formal instrument, in other words, may earn its keep less by preventing a failure than by making the failure cheaper to fix once it happens.

Limitations

Method was not randomly assigned, so any unmeasured difference between organisers who choose an app and organisers who don't could in principle drive part of the incidence gap — though three of the twenty teams had used the other method the previous year, and within those three the channel composition of any leak tracked their **current** year's method rather than their prior year's, suggesting the pattern travels with the mechanism rather than the team. Leak status rests on retrospective self-report against a single confidence threshold, which will misclassify participants near the boundary in both directions equally. All twenty teams were drawn from workplaces in a single country during a single holiday period; the consistency of the channel split across eleven independently organised paper-slip teams and nine indepen-

dently configured app instances suggests the finding is not an artefact of any one team's culture.

Conclusion

An anonymous matching app does not obviously make a gift exchange better kept than a folded slip in a hat; it makes it kept differently, and easier to re-secure once it isn't. Future work might introduce a third draw method — disclosure to a neutral third party who holds every assignment without matching software or a visible slip — to see whether a channel this study didn't test absorbs the leak instead, or whether some baseline leak rate is simply what a small group asked to keep a secret from itself produces regardless of the instrument.

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